12.6 Surface Areas and Volumes of Spheres

Name _____ Period ____ Date _____

Geometry 12.6 · Student sheet

I can…

- find the surface area of a sphere.
- find the volume of a sphere.
- find the surface area and volume of a hemisphere.

Vocabulary · fill in the missing word

Sphere — the set of all points in space that are the same _____ from a given point, the center

Great circle — the _____ of a sphere and a plane that contains the center of the sphere

Hemisphere — half of a sphere, formed when a great circle _____ a sphere into two congruent halves

Part 1 · Discovery: peel an orange

Measure the orange's circumference and find its radius. Trace four circles with that radius. Peel the orange and fill the circles.

Circumference	Radius (C ÷ 2π)	Area of one circle (πr^2)	Circles the peel filled

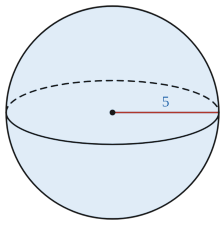
The surface area of a sphere is about _____ times πr^2.

Part 2 · Notes and examples (fill in each note, then work the example)

Surface area of a sphere: S = _____

Example 1

Find the surface area of the sphere. Round to the nearest hundredth.



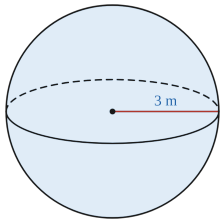
Example 2

A sphere has a surface area of 144π square inches. Find its radius.

Volume of a sphere: V = _____

Example 3

Find the volume of a sphere with a radius of 3 meters. Round to the nearest hundredth.



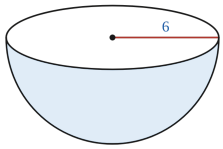
Example 4

A sphere has a surface area of 64π square feet. Find its volume. Round to the nearest hundredth.

Hemisphere: the volume is half the sphere's; the surface area is half of $4\pi r^2$ _____ the flat circle, πr^2.

Example 5

Find the volume and the surface area of the hemisphere. Round to the nearest hundredth.



Part 3 · Partner practice

Partner 1

A ball has a diameter of 10 inches. Find its volume. Round to the nearest hundredth.

Partner 2

Find the surface area of a sphere with a radius of 7. Round to the nearest hundredth.